

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

**Keep in Mind**

**This is a practice test.** It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

**Part A — Vocabulary (8 pts)**

Write the letter of the best description next to each term.

- |                                 |                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1. Subspace <b>F</b>            | (A) The number of vectors in a basis (e.g. $r$ for $C(A)$ , $n - r$ for $N(A)$ ).                               |
| 2. Nullspace $N(A)$ <b>D</b>    | (B) The span of the columns of $A$ ; the reachable outputs $\mathbf{b}$ , a subspace of $\mathbb{R}^m$ .        |
| 3. Column space $C(A)$ <b>B</b> | (C) A set in which only the all-zero combination gives $\mathbf{0}$ (no vector is a combination of the others). |
| 4. Special solution <b>H</b>    | (D) The set of all solutions to $A\mathbf{x} = \mathbf{0}$ ; a subspace of $\mathbb{R}^n$ .                     |
| 5. Linear independence <b>C</b> | (E) The number of pivots $r$ ; equals both the row rank and the column rank.                                    |
| 6. Basis <b>G</b>               | (F) A subset closed under addition and scalar multiplication (so it contains $\mathbf{0}$ ).                    |
| 7. Dimension <b>A</b>           | (G) An independent set that spans the space — the fewest vectors that build every vector in it.                 |
| 8. Rank <b>E</b>                | (H) A solution of $A\mathbf{x} = \mathbf{0}$ found by setting one free variable to 1 and the rest to 0.         |

**Part B — Multiple Choice (12 pts)**

Circle the best answer.

- Which set is **not** a subspace of  $\mathbb{R}^2$ ? (A) the line  $y = 2x$  (B) all of  $\mathbb{R}^2$  **(C) the line  $y = 2x + 1$  ←** (D)  $\{\mathbf{0}\}$
- For an  $m \times n$  matrix of rank  $r$ , the nullspace  $N(A)$  has dimension (A)  $r$  **(B)  $n - r$  ←** (C)  $m - r$  (D)  $n$
- The columns of  $A$  are linearly independent exactly when **(A)  $N(A) = \{\mathbf{0}\}$  ←** (B)  $C(A) = \mathbb{R}^m$  (C)  $A$  is square (D)  $A$  has a zero row
- For *any* matrix, the row rank and the column rank are **(A) always equal ←** (B) equal only if  $A$  is square (C) never equal (D) unrelated
- When  $A\mathbf{x} = \mathbf{b}$  is solvable, its complete solution has the form (A)  $\mathbf{x}_p$  only (B)  $\mathbf{x}_n$  only **(C)  $\mathbf{x}_p + \mathbf{x}_n$  (one particular + the whole nullspace) ←** (D) always a single point
- If  $\mathbf{x}$  is in the nullspace of  $A$ , then  $\mathbf{x}$  is perpendicular to (A) every column of  $A$  **(B) every row of  $A$  ←** (C) the vector  $\mathbf{b}$  (D) nothing

Teacher Note

Item 1: a subspace must contain  $\mathbf{0}$ ;  $y = 2x + 1$  misses the origin (and its points don't add to a point on the line).  
 Item 3: independent columns  $\iff$  the only solution of  $A\mathbf{x} = \mathbf{0}$  is  $\mathbf{x} = \mathbf{0}$ . Item 6:  $A\mathbf{x} = \mathbf{0}$  read row by row says (row)  $\cdot \mathbf{x} = 0$  for every row — the Part D / FTLA idea.

Part C — Short Answer & Computation (35 pts)

Show your work. Problems 2–7 all use the same matrix  $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 5 \end{bmatrix}$ .

- Use the **closure test** to decide whether  $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$  is a subspace of  $\mathbb{R}^3$ . **Yes —  $W$  is a subspace (a plane through the origin).** **Contains  $\mathbf{0}$ :**  $0 + 0 + 0 = 0$ . **Addition:** if  $(x_1 + y_1 + z_1) = 0$  and  $(x_2 + y_2 + z_2) = 0$ , adding gives  $(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = 0$ , so the sum is in  $W$ . **Scaling:** multiplying by  $c$  gives  $c(x + y + z) = c \cdot 0 = 0$ , still in  $W$ . All three hold.
- Row-reduce  $A$  and find every **special solution** of  $A\mathbf{x} = \mathbf{0}$ . State  $\dim N(A)$ .  $R_2 - 2R_1$ ,  $R_3 - 3R_1$  then  $R_3 - R_2$  give the reduced form  $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ . **Columns 1, 2 are pivots; column 3 is free.** Set  $x_3 = 1$ :  $x_1 = -1$ ,  $x_2 = -1$ , so the special solution is  $\mathbf{s} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ . **One free variable, so  $\dim N(A) = n - r = 3 - 2 = 1$ .**
- Solve  $A\mathbf{x} = \mathbf{b}$  completely for  $\mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ . **Reducing  $[A \mid \mathbf{b}]$  gives  $\begin{bmatrix} 1 & 0 & 1 & | & 1 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$  (consistent).** Free  $x_3 = 0$  gives a particular solution  $\mathbf{x}_p = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ . **Complete solution:**  $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ .
- Is  $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  in the column space  $C(A)$ ? **No.** Reducing  $[A \mid \mathbf{b}]$  gives a last row  $[0 \ 0 \ 0 \mid 1]$ , i.e.  $0 = 1$  — impossible. The system is inconsistent, so  $\mathbf{b} \notin C(A)$ .
- Are the columns of  $A$  linearly independent? Give a **basis** for  $C(A)$  and its dimension. **Not independent —  $N(A) \neq \{\mathbf{0}\}$ , and  $\mathbf{s} = (-1, -1, 1)$  says column 3 = column 1 + column 2:**  $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ . **The pivot columns form a basis:**  $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}$ , so  $\dim C(A) = r = 2$ .
- State the dimensions of all four subspaces and where each lives.  $\dim C(A) = r = 2$ ,  $\dim N(A) = n - r = 1$ ,  $\dim C(A^T) = r = 2$  (row space),  $\dim N(A^T) = m - r = 1$  (left nullspace). **Input  $\mathbb{R}^3$ :** row space  $C(A^T)$  and  $N(A)$  ( $2 + 1 = 3$ ). **Output  $\mathbb{R}^3$ :** column space  $C(A)$  and left nullspace  $N(A^T)$  ( $2 + 1 = 3$ ).
- Verify that  $\mathbf{s} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$  is perpendicular to every row of  $A$ . **Row 1:**  $(1, 1, 2) \cdot (-1, -1, 1) = -1 - 1 + 2 = 0$ . **Row 2:**  $(2, 1, 3) \cdot \mathbf{s} = -2 - 1 + 3 = 0$ . **Row 3:**  $(3, 2, 5) \cdot \mathbf{s} = -3 - 2 + 5 = 0$ . **All zero, so  $\mathbf{s} \perp$  every row. This illustrates the Fundamental Theorem of Linear Algebra:  $N(A)$  is orthogonal to the row space  $C(A^T)$ .**

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- Consider the line  $W = \{(x, y) \in \mathbb{R}^2 : y = 2x + 1\}$ . (a) **No.** Putting  $(x, y) = (0, 0)$  gives  $0 = 2(0) + 1 = 1$ , which is false, so  $\mathbf{0} \notin W$ . (b)  $W$  is *not* a subspace. A subspace must contain  $\mathbf{0}$ , and  $W$  fails that at once; closure also fails (e.g.  $(0, 1)$  and  $(1, 3)$  are on the line but their sum  $(1, 4)$  is not, since  $4 \neq 2(1) + 1 = 3$ ). (c) The line  $y = 2x$  *is* a subspace: it passes through the origin and equals all multiples  $t(1, 2)$ , so it is closed under addition and scaling. The “+1” shifts the line off the origin and breaks every closure property.

2. For  $A$  above (rank 2, special solution  $\mathbf{s} = (-1, -1, 1)$ ): (a)  $A\mathbf{x} = \mathbf{0}$  read one row at a time says (row 1)  $\cdot \mathbf{x} = 0$ , (row 2)  $\cdot \mathbf{x} = 0$ , (row 3)  $\cdot \mathbf{x} = 0$ . A zero dot product means perpendicular (Lesson 1.2), so any  $\mathbf{x}$  in  $N(A)$  is perpendicular to every row — hence to their whole span, the row space. (b) The dimensions add to  $n$ :  $\dim N(A) + \dim C(A^T) = 1 + 2 = 3$ , so together they have enough directions to reach every vector in  $\mathbb{R}^3$ . They meet only at  $\mathbf{0}$  because a vector  $\mathbf{v}$  in *both* would be perpendicular to itself, giving  $\mathbf{v} \cdot \mathbf{v} = 0$ , which forces  $\mathbf{v} = \mathbf{0}$ . Perpendicular + dimensions filling the space  $\Rightarrow$  they are orthogonal complements.

Teacher Note

**Scoring Part D (5 pts each).** D1: 1 pt (a)  $\mathbf{0} \notin W$  with the  $0 = 1$  check; 2 pts (b) “not a subspace” *because* it misses  $\mathbf{0}$  / fails closure; 2 pts (c)  $y = 2x$  is a subspace and *why* (through origin, closed). D2: 3 pts (a) the row-by-row argument (dot = 0  $\Rightarrow \perp$ , so  $N(A) \perp$  row space); 2 pts (b) dims  $1 + 2 = 3$  fill  $\mathbb{R}^3$  and  $\mathbf{v} \cdot \mathbf{v} = 0 \Rightarrow \mathbf{v} = \mathbf{0}$  for the  $\{\mathbf{0}\}$  intersection. Accept “orthogonal complements” as the summary.